

Consecutive-prime Artin correlations across squarefree bases: a census, a residue-channel reconstruction, and lagged cross-base correlations

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Abstract

This data note records freshly computed primitive-root indicators on consecutive prime pairs. A census over the 64 squarefree bases from 2 through 105, with $7 \leq p < q < 1\,000\,000\,000$, gives 16 positive consecutive-pair associations; the remainder are negative. For base 15, extending the computation to $10\,000\,000\,000$ gives $\delta = +0.004646$, compared with $+0.012196$ at the smaller bound. For a selected set of 12 bases, we release the full lagged cross-base matrix and reconstruct the same-base associations by combining conditional Artin progression densities with measured residue-gap channel frequencies. The ratio of reconstructed to empirical association lies between 0.969 and 1.040. These frequencies are inputs: this is an empirical-weight consistency check, not an independent prediction. An elementary quadratic-character argument gives a worked example of a mixed-base obstruction for bases 2 and 6 in either order at gaps congruent to 10 or 14 modulo 24. The numerical results are computations; the displayed obstruction is an elementary proof within an existing framework. No asymptotic claim is made.

1 Scope and relationship to earlier work

For an integer base a and a prime p , write $\mathcal{A}_a(p)$ for the indicator that a is a primitive root modulo p ; it is zero when p divides a . Artin’s conjecture concerns the density of primes for which this indicator is one. Hooley proved the expected density under the relevant generalized Riemann hypotheses (GRH) [1]; Moree surveys the density formulas, their entanglement corrections, and primitive roots in progressions [2]. Here the question is finite and empirical: how do the indicators at adjacent primes vary across bases, and how much of their association is reproduced by single-prime progression rates combined with measured prime transitions?

The distinction between a single-prime density and a consecutive-pair law is essential. Lemke Oliver and Soundararajan give a conjectural quantitative framework for biases in successive reduced residue classes [3]. Their Hardy–Littlewood-based framework does not make the present measured weights into theoretical inputs. Nor does GRH for the individual Artin densities supply conditional independence for adjacent prime indicators.

Bald’s consolidated manuscript [6] treats the base-10 consecutive-prime results, their exact residue decomposition, same-prime cross-base correlations, and computational usefulness. We refer there rather than reproduce those analyses. Bald’s quadratic exclusion note [5] provides the same-base setting. The additions here are a broader squarefree-base census, a uniform multi-base reconstruction, and a matrix in which the two bases are attached to *different consecutive primes*. These are dataset extensions, not a proposed theory of correlations.

Tinková, Waxman and Zindulka study Artin twin primes [4]. Their discussion already allows distinct primitive-root bases at distinct shifted primes. Thus neither the mixed-base framework nor the necessary quadratic nonresidue condition is claimed as new. The small worked example below is included to make the arithmetic behind particular matrix entries transparent.

2 Definitions and exact finite conventions

For a cutoff x , let $\mathcal{P}_x$ consist of consecutive rational prime pairs (p, q) with $7 \leq p < q < x$. Consecutiveness is in the full rational prime sequence, not in a subsequence selected by primitive-root status or by a residue condition. Both endpoints must be below the bound. Give all pairs equal weight. For a source base a and a destination base b , put

$$X = \mathcal{A}_a(p), \quad Y = \mathcal{A}_b(q), \quad n_{ij} = \#\{(p, q) \in \mathcal{P}_x : X = i, Y = j\}.$$

Our directional association is

$$\begin{aligned} \delta(a \rightarrow b; x) &= \Pr(Y = 1 \mid X = 1) - \Pr(Y = 1 \mid X = 0) \\ &= \frac{n_{11}}{n_{10} + n_{11}} - \frac{n_{01}}{n_{00} + n_{01}} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}. \end{aligned} \tag{1}$$

The marginals are nondegenerate in every reported table. For $a = b$, abbreviate this to $\delta_a(x)$. The normalization uses the *source* variance, not a symmetric Pearson normalization. In particular, the difference between the two orders in the matrix can involve both covariance and marginal normalization.

Adjacent pairs overlap at a prime. We report exact finite counts and descriptive summaries only. There are no independent-pair error bars, null-test conclusions, or claims that a descriptive score certifies a sign. The two cutoffs for base 15 are nested cumulative samples, not independent replications.

The census bases are precisely the squarefree integers in $[2, 105]$; this is not the set of all nonsquare integers in that interval. The selected matrix and reconstruction bases are

$$\mathcal{B} = \{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}.$$

For squarefree positive a , the fundamental discriminant and quadratic conductor are

$$D_a = \begin{cases} a, & a \equiv 1 \pmod{4}, \\ 4a, & \text{otherwise,} \end{cases} \quad \chi_a(r) = \left(\frac{D_a}{r} \right).$$

The symbol is the Kronecker symbol, and on odd primes not dividing a it agrees with the Legendre symbol of a . This convention covers the even conductors without treating the character as a Legendre symbol at a composite denominator.

3 The squarefree-base census

There are 50 847 530 pairs in the main census. Table 1 gives all contingency counts, conductors and associations, allowing the displayed decimal values to be checked without rerunning a prime sieve. Exactly 16 of the 64 associations are positive and 48 are negative. The positive bases are

$$\{15, 26, 30, 35, 38, 39, 42, 51, 55, 66, 70, 78, 87, 95, 102, 105\}.$$

Within this finite set their minimum quadratic conductor is 60. This is a description of the recorded sample, not a sign criterion or a rule for bases outside it. No fitted character statistic or sign predictor is used.

The positive values show why base-10 anticorrelation should not be silently extended to every base at the same bound. Conversely, positivity at this bound says nothing about an eventual sign. A conductor is useful metadata, not by itself an explanation of the ordering of all the measured values.

For base 15, the fresh extension has 455 052 507 pairs and gives

$$\delta_{15}(1\,000\,000\,000) = +0.012196, \quad \delta_{15}(10\,000\,000\,000) = +0.004646.$$

It remains positive and is smaller at the larger cutoff. These two values do not determine a decay rate or limiting behavior. In particular, treating the larger nested sample as a new independent positive-sign experiment would misrepresent how the data were obtained.

Table 1: Squarefree-base census. All consecutive pairs have $7 \leq p < q < 1\,000\,000\,000$. Counts use the order (X, Y) ; D_a is the quadratic conductor.

a	D_a	n_{00}	n_{01}	n_{10}	n_{11}	δ_a
2	8	19 331 100	12 501 645	12 501 645	6 513 140	-0.050199
3	12	19 373 402	12 459 637	12 459 638	6 554 853	-0.046676
5	5	17 887 263	12 944 550	12 944 551	7 071 166	-0.066563
6	24	19 630 351	12 203 765	12 203 765	6 809 649	-0.025205
7	28	19 766 161	12 065 063	12 065 063	6 951 243	-0.013491
10	40	19 758 045	12 072 868	12 072 869	6 943 748	-0.014140
11	44	19 700 431	12 129 815	12 129 814	6 887 470	-0.018909
13	13	19 267 755	12 442 700	12 442 700	6 694 375	-0.042573
14	56	19 839 940	11 992 198	11 992 198	7 023 194	-0.007390
15	60	20 071 423	11 759 377	11 759 377	7 257 353	+0.012196
17	17	19 445 115	12 315 810	12 315 811	6 770 794	-0.033025
19	76	19 879 965	11 951 632	11 951 633	7 064 300	-0.003971
21	21	19 586 383	12 336 717	12 336 717	6 587 713	-0.038345
22	88	19 862 731	11 969 812	11 969 811	7 045 176	-0.005518
23	92	19 891 024	11 939 021	11 939 021	7 078 464	-0.002878
26	104	19 948 674	11 884 698	11 884 699	7 129 459	+0.001614
29	29	19 630 954	12 177 294	12 177 294	6 861 988	-0.022422
30	120	20 244 448	11 589 080	11 589 080	7 424 922	+0.026445
31	124	19 906 368	11 926 243	11 926 243	7 088 676	-0.001859
33	33	19 755 228	12 111 764	12 111 764	6 868 774	-0.018187
34	136	19 897 813	11 932 677	11 932 677	7 084 363	-0.002355
35	140	19 980 168	11 850 522	11 850 522	7 166 318	+0.004542
37	37	19 656 884	12 159 789	12 159 789	6 871 068	-0.021134
38	152	19 934 198	11 896 013	11 896 014	7 121 305	+0.000731
39	156	20 065 680	11 764 518	11 764 518	7 252 814	+0.011777
41	41	19 695 440	12 122 551	12 122 551	6 906 988	-0.018035
42	168	20 038 405	11 794 912	11 794 912	7 219 301	+0.009158
43	172	19 910 105	11 920 219	11 920 219	7 096 987	-0.001305
46	184	19 904 623	11 925 964	11 925 964	7 090 979	-0.001793
47	188	19 917 453	11 915 193	11 915 193	7 099 691	-0.000932
51	204	20 005 626	11 827 036	11 827 036	7 187 832	+0.006473
53	53	19 733 025	12 092 997	12 092 996	6 928 512	-0.015726

Continued on next page

a	D_a	n_{00}	n_{01}	n_{10}	n_{11}	δ_a
55	220	20 045 243	11 785 063	11 785 063	7 232 161	+0.010049
57	57	19 863 814	11 977 842	11 977 842	7 028 032	-0.006387
58	232	19 880 001	11 951 635	11 951 635	7 064 259	-0.003972
59	236	19 870 776	11 963 818	11 963 819	7 049 117	-0.005058
61	61	19 765 583	12 059 798	12 059 799	6 962 350	-0.012924
62	248	19 930 390	11 904 047	11 904 046	7 109 047	-0.000033
65	65	19 899 839	11 937 009	11 937 008	7 073 674	-0.002854
66	264	20 023 849	11 807 704	11 807 704	7 208 273	+0.008121
67	268	19 907 419	11 925 775	11 925 775	7 088 561	-0.001832
69	69	19 855 663	11 981 295	11 981 295	7 029 277	-0.006577
70	280	20 050 133	11 781 978	11 781 978	7 233 441	+0.010270
71	284	19 906 892	11 927 214	11 927 214	7 086 210	-0.001973
73	73	19 800 404	12 026 529	12 026 529	6 994 068	-0.010162
74	296	19 921 196	11 913 394	11 913 394	7 099 546	-0.000822
77	77	19 877 215	11 955 403	11 955 403	7 059 509	-0.004309
78	312	19 940 372	11 892 936	11 892 936	7 121 286	+0.000924
79	316	19 897 841	11 933 990	11 933 990	7 081 709	-0.002494
82	328	19 877 198	11 956 924	11 956 924	7 056 484	-0.004469
83	332	19 891 316	11 937 178	11 937 178	7 081 858	-0.002691
85	85	19 926 792	11 906 935	11 906 935	7 106 868	-0.000261
86	344	19 901 808	11 931 119	11 931 119	7 083 484	-0.002276
87	348	19 969 919	11 861 682	11 861 683	7 154 246	+0.003585
89	89	19 804 042	12 025 963	12 025 963	6 991 562	-0.010181
91	364	19 915 387	11 916 900	11 916 899	7 098 344	-0.001068
93	93	19 811 004	12 026 822	12 026 821	6 982 883	-0.010420
94	376	19 907 194	11 920 838	11 920 838	7 098 660	-0.001308
95	380	19 969 782	11 863 309	11 863 309	7 151 130	+0.003417
97	97	19 827 544	12 003 768	12 003 768	7 012 450	-0.008344
101	101	19 824 663	12 006 863	12 006 864	7 009 140	-0.008609
102	408	19 933 636	11 899 037	11 899 037	7 115 820	+0.000425
103	412	19 911 006	11 924 104	11 924 105	7 088 315	-0.001733
105	105	20 055 906	11 771 510	11 771 510	7 248 604	+0.011248

4 An empirical-weight residue-channel reconstruction

4.1 Inputs and single-prime rates

For each $a \in \mathcal{B}$, set $M_a = \text{lcm}(D_a, 840)$. A channel is the pair (r, g) , where $r = p \bmod M_a$ and $g = q - p$ is the *exact* gap. Its measured weight is

$$w_a(r, g) = \frac{\#\{(p, q) \in \mathcal{P}_x : p \equiv r \pmod{M_a}, q - p = g\}}{|\mathcal{P}_x|}.$$

All channel frequencies are computed from the same prime stream used for the census. They do not use the primitive-root labels, but they do use the measured successor distribution. They must therefore be counted as empirical inputs.

For the squarefree bases considered here, with the conductor contained in the modulus, the GRH Artin progression-rate expression can be written as

$$\alpha_a(r) = \mathbf{1}_{(r, M_a)=1} \mathbf{1}_{\chi_a(r)=-1} \prod_{\substack{\ell | M_a, \ell > 2 \\ r \not\equiv 1 \pmod{\ell}}} \left(1 - \frac{1}{\ell}\right) \prod_{\ell \nmid M_a} \left(1 - \frac{1}{\ell(\ell-1)}\right). \quad (2)$$

Both products run over primes. The first product describes the odd-power obstructions whose congruence condition is already fixed by r ; the quadratic condition is written separately. The remaining Euler factors average the unfixed obstructions. This is the squarefree-base, conductor-containing case of the classical Artin progression formulas, not a formula asserted for arbitrary perfect-power bases or arbitrary moduli [2]. Averaging (2) over the reduced classes modulo M_a recovers Hooley’s density for base a . This includes the entanglement correction for $a \equiv 1 \pmod{4}$, where $\mathbb{Q}(\sqrt{a}) \subset \mathbb{Q}(\zeta_a)$. Because the conductor divides M_a , the restriction $\chi_a(r) = -1$ in (2) encodes it.

We evaluate the Euler product through 10 000 000, accumulating with 40 decimal digits. That arithmetic precision is not a claim about the infinite product. If A_B is the truncated Artin product, then

$$A_B \left(1 - \frac{1}{B}\right) \leq A \leq A_B, \quad B = 10\,000\,000,$$

since the sum of the omitted factors’ deficits is at most $\sum_{n>B} 1/[n(n-1)] = 1/B$. Thus a conservative bound for the relative omitted-factor error is 0.0000001. The saved output contains both bounds, and the reconstruction propagates them. The quoted ratio range uses only 3 decimal places; fine agreement should not be read beyond the precision justified by the product and by the model itself.

4.2 The contraction, and what it does not test

Model I replaces the two endpoint statuses, within a channel, by independent Bernoulli variables having rates $\alpha_a(r)$ and $\alpha_a(r+g)$. Define

$$\begin{aligned} u_a &= \sum_{r,g} w_a(r,g) \alpha_a(r), & v_a &= \sum_{r,g} w_a(r,g) \alpha_a(r+g), \\ t_a &= \sum_{r,g} w_a(r,g) \alpha_a(r) \alpha_a(r+g), & \delta_{a,\text{rec}} &= \frac{t_a - u_a v_a}{u_a(1 - u_a)}. \end{aligned}$$

The argument of α_a is reduced modulo M_a . No empirical Artin marginal is substituted into this normalization, and no fitted scale or intercept is used. The empirical comparison is exactly (1) on the same pair set.

Table 2 gives a ratio range from 0.969 to 1.040. The positive base-15 entry is reconstructed as positive, while the other displayed same-base associations are negative. This agreement is useful as a consistency check that the measured prime-transition distribution and classical single-prime progression rates can account for much of the finite-range pattern across these bases.

It is not a Hardy–Littlewood derivation: the gap and residue weights were not predicted. It is not a test establishing conditional independence: that independence was imposed in forming the product of rates. A small aggregate error can conceal compensating within-channel errors. Nor does a good aggregate fit bound information carried by the predecessor after conditioning; that is a different question treated for base 10 in [6].

Although the raw weights retain exact gaps for transparency, this particular contraction uses a gap only through the destination residue. Summing weights with the same $(r, r+g \bmod M_a)$ leaves the reconstruction unchanged. The exact-gap representation is therefore a data organization, not an additional independence mechanism.

4.3 Finite nonreduced-prime exceptions

Formula (2) assigns zero to nonreduced classes. This is an asymptotic progression convention: a prime dividing the modulus can still be an Artin prime for the base. In the reported pair range,

a	M_a	δ_{emp}	δ_{rec}	Ratio
2	840	-0.050199	-0.049831	0.9927
3	840	-0.046676	-0.047998	1.0283
5	840	-0.066563	-0.066471	0.9986
6	840	-0.025205	-0.024430	0.9692
7	840	-0.013491	-0.013454	0.9973
10	840	-0.014140	-0.014703	1.0398
11	9 240	-0.018909	-0.018977	1.0036
13	10 920	-0.042573	-0.042830	1.0060
15	840	+0.012196	+0.012159	0.9970
17	14 280	-0.033025	-0.033055	1.0009
21	840	-0.038345	-0.038245	0.9974
29	24 360	-0.022422	-0.022473	1.0023

Table 2: Model I with empirical residue-gap weights. Ratio means $\delta_{a,\text{rec}}/\delta_{a,\text{emp}}$, not predictive accuracy on a held-out sample.

the channel $(r, g) = (7, 4)$ for base 17 contains a doubly-Artin pair, namely $(7, 11)$, even though its reconstructed product rate is zero. The fresh stream logs those exact endpoints and the channel array independently retains their contribution.

For clarity about the lower endpoint, if the range were enlarged to start at 5, the additional pair $(5, 7)$ would be another such exception for base 17 and also an exception for base 3. Direct modular-order checks identify all of these finite events. They are not violations of a character obstruction: they arise from primes dividing the chosen modulus. We do not discard them from the empirical counts or assert that every zero-rate channel is empty.

5 Lagged cross-base correlations

The matrix has source a on rows and destination b on columns. Tables 3 and 4 give all entries; the raw data also retain each contingency table, rather than just the rounded associations. The diagonal agrees with the selected same-base census values. The 132 off-diagonal entries contain 101 negative values and have mean -0.003454 .

Across unordered pairs of distinct bases, the mean absolute difference between opposite directions is 0.008958, with maximum 0.051099 at $\{6, 10\}$. These are descriptive summaries of the directional statistic. The matrix is not the same-prime cross-base matrix: $\mathcal{A}_a(p)$ and $\mathcal{A}_b(q)$ are measured at different primes, whose residue classes have an oriented successor distribution. The same-prime entanglement discussion in [6] cannot simply be relabeled to explain these signs.

In particular, an exclusion of joint successes in some gap classes does not force the global association to be negative. The global quantity also combines all other classes and the source/destination marginals. The mixed pair studied next illustrates a local necessary-condition obstruction, not a global sign law for the matrix. We make no classification of matrix signs in terms of multiplicative relations among bases.

$a \backslash b$	2	3	5	6	7	10
2	-0.050199	-0.003005	-0.014488	+0.009426	-0.002789	-0.010957
3	-0.002725	-0.046676	-0.014747	-0.002801	-0.023345	-0.003055
5	+0.003313	+0.003365	-0.066563	+0.003188	+0.003217	+0.003329
6	-0.035420	-0.002852	-0.014710	-0.025205	-0.002883	+0.041485
7	-0.002899	-0.019289	-0.014534	-0.002819	-0.013491	-0.002846
10	-0.002569	-0.002930	-0.014715	-0.009614	-0.002842	-0.014140
11	-0.002527	-0.020131	-0.014342	-0.002732	-0.003744	-0.002789
13	-0.000085	-0.000135	-0.010909	+0.000047	+0.000054	-0.000118
15	-0.002858	+0.022208	-0.014599	-0.002823	+0.001448	-0.002940
17	-0.004351	-0.004503	-0.017752	-0.004346	-0.004818	-0.004413
21	-0.010183	-0.010350	-0.033040	-0.009939	-0.010200	-0.009731
29	-0.001001	-0.001568	-0.008987	-0.001271	-0.001206	-0.001472

Table 3: Lagged matrix, first block of destination bases. Entries are $\delta(a \rightarrow b; 1\,000\,000\,000)$.

$a \backslash b$	11	13	15	17	21	29
2	-0.002754	-0.006972	-0.002838	-0.002966	+0.004450	-0.003509
3	-0.003210	-0.007078	+0.031438	-0.002868	+0.004336	-0.003712
5	+0.003522	-0.000430	+0.003255	+0.000538	+0.002513	+0.006611
6	-0.002943	-0.007193	-0.002921	-0.002932	+0.004245	-0.003622
7	-0.002139	-0.007268	+0.006698	-0.002756	+0.004127	-0.003728
10	-0.002887	-0.007276	-0.002888	-0.002887	+0.004185	-0.003521
11	-0.018909	-0.007341	-0.000822	-0.003006	+0.004503	-0.003595
13	+0.000053	-0.042573	+0.000117	+0.000048	+0.003391	-0.001002
15	-0.000801	-0.007280	+0.012196	-0.002629	+0.004258	-0.003518
17	-0.004325	-0.008888	-0.004493	-0.033025	+0.001414	-0.005485
21	-0.009991	-0.016546	-0.010018	-0.008659	-0.038345	-0.012848
29	-0.001185	-0.005830	-0.001066	-0.001515	+0.004144	-0.022422

Table 4: Lagged matrix, remaining destination bases. Together with Table 3, this is the full square matrix.

6 A worked mixed quadratic obstruction

For bases a, b , let $L = \text{lcm}(2, D_a, D_b)$. An even shift $g \bmod L$ is called *nonvacuously excluded* when there is at least one residue r for which both r and $r + g$ are coprime to L , but no such residue with

$$\chi_a(r) = \chi_b(r + g) = -1.$$

The nonvacuity restriction matters: an odd shift modulo an even modulus cannot join two reduced residues, and must not be counted as an informative exclusion. These conditions are necessary quadratic conditions for simultaneous primitive roots, not sufficient conditions for their existence in all remaining classes.

Proposition 1. *Let $p, q > 3$ be primes, $q - p \equiv 10$ or $14 \pmod{24}$. Then it is impossible for 2 to be a primitive root modulo p and 6 to be a primitive root modulo q . The same statement holds with the bases interchanged. Consecutiveness is not needed.*

Proof. For a prime greater than 3, the reduced residues modulo 24 on which $\chi_2 = -1$ are 5, 11, 13, 19. Adding either specified gap gives the finite table below. Whenever the successor residue is reduced, its χ_6 value is $+1$, so 6 cannot be a primitive root there. If it is not reduced, it cannot be the residue of the stipulated prime q . This proves the first order. For the reversed order, apply the same residue argument to the reversed endpoint residues: negation of the gap interchanges 10 and 14 modulo 24. The argument depends only on residues, not on which endpoint is larger. $\square$

g	r	$r + g \pmod{24}$	Successor condition
10	5	15	not a reduced residue
10	11	21	not a reduced residue
10	13	23	$\chi_6 = +1$
10	19	5	$\chi_6 = +1$
14	5	19	$\chi_6 = +1$
14	11	1	$\chi_6 = +1$
14	13	3	not a reduced residue
14	19	9	not a reduced residue

Table 5: The complete residue check used in the mixed obstruction.

We enumerate all ordered pairs in $\mathcal{B}$ directly from the character values over residues. The only off-diagonal pairs with nonvacuous excluded even classes are $(2, 6)$ and $(6, 2)$, each with the two classes in the proposition. Exactly 9 diagonal pairs have classes; the full diagonal scan is in Table 6. This is an exhaustive statement about the selected finite set, not a classification for arbitrary bases.

In the main prime census, 7 604 098 pairs have one of the two specified gap classes, and there are 0 simultaneous successes in either order. Separately, a deliberately naive primitive-root implementation enumerates multiplicative orders for both bases at every prime in the range through 1 000 000. Among 78 494 consecutive pairs it finds 12 278 with the specified classes and 0 violations in either order. This check validates the implementation against the elementary argument; the finite computation is not the proof of the all-primes statement.

The proposition is also formally verified in the Lean proof assistant with Mathlib. The package directory `lean/` states both orders as `MixedExclusion.not_primitiveRoot_two_six_of_gap` and `MixedExclusion.not_primitiveRoot_six_two_of_gap`, with the gap condition written as a

Base	Modulus	Nonvacuous excluded even classes
2	8	4
3	12	4,6,8
5	10	2,8
6	24	8,12,16
7	28	14
10	40	20
11	44	22
13	26	none
15	60	20,30,40
17	34	none
21	42	14,28
29	58	none

Table 6: Diagonal part of the finite character scan. An empty set is recorded as none; it does not assert existence of Artin pairs in every shift.

congruence of residues modulo 24 and primitive roots expressed through Mathlib’s `IsPrimitiveRoot`. The proof uses Euler’s criterion, the second supplementary law, quadratic reciprocity for 3 and an exhaustive residue check. The checked statements depend only on Lean’s standard axioms, and the build contains no unproved steps. The formalization covers this proposition only, not the computations.

7 Computation, validation and reproducibility

All large enumerations and the residue/reconstruction calculations were run fresh on the author’s multi-core workstation. The sieve source was transferred as exact base-encoded bytes and its cryptographic checksum checked before compilation. The package records source checksums, commands, output checksums and runtimes. The main runs used 28 OpenMP threads, with approximate census runtimes 41.63 seconds for the multi-base bound and 38.67 seconds for the single-base extension. These are machine-specific execution records, not algorithmic speed comparisons.

The C census marks composites in independent half-open integer blocks. Prime labels are computed within blocks and the resulting streams are assembled in block order, retaining the last prime and its labels across each boundary. Thus no halo length or assumed maximum prime gap is needed to recover consecutiveness. Only the initial integers below the prime range are specially excluded; the first integers of later blocks are sieved normally.

For each prime, distinct factors of $p - 1$ are obtained once. For every base, the primitive-root test checks that the base is a unit and that $a^{(p-1)/\ell} \not\equiv 1 \pmod{p}$ for every prime divisor ℓ of $p - 1$. Modular multiplication uses unsigned double-width integer arithmetic. The single-base extension uses the same prime-stream construction and root test. The gap arrays have 150 slots for even gaps up to 300; there is an assertion, not silent overflow clamping. The observed maximum gap in the main run is 282.

Before scaling, an independent Python implementation trial-divided integers for primality and enumerated complete multiplicative orbits through 1000. For block sizes 2, 17, 31, 1000, it checked every prime label, every same-base and mixed-base count, and every nonzero channel cell. This includes boundaries deliberately placed inside the small prime range. The same tiny computation on

the workstation agreed exactly. A separate C orbit-enumeration program, checked on the tiny range, supplies the naive mixed-law check; it does not reuse the factor-of- $p - 1$ criterion.

For the character calculation, the Kronecker routine was checked against Euler’s criterion on small odd primes. The exclusion scan represents reduced residue sets and negative-character sets as bitsets and intersects cyclic shifts. It enumerates every even shift and tests nonvacuity before testing exclusion. The reconstruction reads the saved channel arrays and applies the printed formulas without an empirical primitive-root-rate substitution.

The numerical TeX macros, tables and bibliographic identifiers are generated by the script `code/make_tables.py`, using saved outputs and explicit metadata. The raw binary channel files include their base, modulus, gap-slot count and pair-count headers; their format is documented in the README. All final numerical displays can therefore be rebuilt without a fresh sieve, while the source needed for a full resieve is also supplied. Exact counts are primary; rounding is confined to generated presentation material.

8 Interpretation and limitations

The note adds a portable finite census and a multi-base comparison, not a new mechanism beyond the residue-channel perspective. The same-base positive examples prevent an overgeneralization of the base-10 behavior. The lagged matrix records an observable different from same-prime entanglement. The reconstruction separates measured prime-transition frequencies from classical single-prime rates, making its empirical inputs explicit.

None of these observations settles the large-cutoff behavior of actual Artin indicators. In the separable fixed-modulus model considered in [6], product limiting residue marginals force the modelled covariance to vanish: the joint rate is the average of a product of separate endpoint functions. The present contraction has exactly that separable form. Unequal finite gap weights do not invalidate that elementary observation. It is a statement about the restricted model under a limiting residue-law assumption, not a proof of an asymptotic for the actual indicators. The finite-range data here do not settle that asymptotic question.

A genuine prediction would require weights obtained without the measured successor stream, and a justified account of any residual endpoint dependence. Increasing a table’s size or matching its aggregate covariance does not provide either ingredient. We regard the present package as a reproducible data and consistency-check supplement to the cited work, with the mixed obstruction as an explanatory example.

Data availability and licences

The reproduction package `reproduction.zip` accompanies this preprint. It contains the raw outputs, source programs, generated tables, manuscript, build instructions and cryptographic checksum manifests. Its unpacked contents are public at <https://github.com/jpbald93/multibase-artin-correlations>. No deposit identifier is asserted here. Text is licensed under CC BY 4.0; code under MIT.

AI assistance

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